# OMNIX QUANTUM

Decision Governance Infrastructure | Post-Quantum Cryptography | EU AI Act Ready

## Executive Verification Package

Runtime Treasury Execution Trace | OMNIX-RTE-001 | v1.4.0

**WHAT THIS PACKAGE PROVES**

An AI agent managing a USD 50,000,000 treasury transaction cannot release funds without (1) valid human authority, (2) a sealed mandate contract formed before any action is taken, and (3) real-time behavioral monitoring throughout execution. Every step of this process is recorded with post-quantum cryptographic seals and is verifiable offline by any independent party -- no access to OMNIX servers required.

| PATH A | PATH B | PATH C |
|---|---|---|
| **BLOCKED** | **APPROVED + PoGC** | **HALTED MID-RUN** |
| Invalid authority | Valid authority + certificate | Mandate drift detected |

**SCENARIO**

Cross-border liquidity release -- EUR counterparty -- USD 50,000,000 via SWIFT MT202 / XRPL RLUSD

| | | | | | |
|---|---|---|---|---|---|
| **Package:** | OMNIX-RTE-001-694384C0419A4715 | **Generated:** | 2026-05-30 20:59 UTC | **Version:** | 1.4.0 |

| | |
|---|---|
| **Cryptography:** | ML-DSA-65 (Dilithium-3, FIPS 204) -- Post-Quantum Standard |
| **Regulatory:** | EU AI Act Art. 9 / MiCA Title VI / DORA Art. 11 |
| **Standards:** | RFC-ATF-1 through RFC-ATF-6 (all published with DOI) |

## THE THREE SCENARIOS -- One Decision Engine, Three Outcomes

The same AI agent, the same USD 50,000,000 treasury transaction, the same governance engine -- three different authority conditions. The package proves that the outcome depends entirely on the integrity of authority and behavior throughout execution, not just at the point of approval.

### PATH A BLOCKED BEFORE EXECUTION -- Invalid Authority Detected

*The AI agent held degraded authority (42%). The governance engine refused before any action was taken.*

| | |
|---|---|
| **Authority level** | 42% -- DEGRADED (recertification required, not completed) |
| **Continuity score** | 42.05 / 100 -- Band: CRITICAL |
| **Governance gate** | OSG (Settlement Gate) returned: REJECTED |
| **Funds released** | USD 0 -- transaction was BLOCKED |
| **Execution occurred** | No -- agent never touched the funds |
| **Cryptographic record** | Full audit trail sealed with ML-DSA-65 post-quantum signature |

**VERDICT: NO EXECUTION -- USD 0 RELEASED -- FUNDS PROTECTED**

### PATH B APPROVED WITH CERTIFICATE -- Recertified Authority, Full Mandate Compliance

*The AI agent held recertified authority (88%). Three-turn execution completed with zero mandate violations.*

| | |
|---|---|
| **Authority level** | 88% -- RECERTIFIED (full re-authorization on record) |
| **Continuity score** | 95.5 / 100 -- Band: NOMINAL |
| **Mandate compliance** | Certification: N/A -- Violations: 0 -- Turns: 4 |
| **Governance gate** | OSG (Settlement Gate) returned: APPROVED |
| **Funds released** | USD 50,000,000 -- XRPL: XRPL-7F572A21E36C47CE925CE0E3 |
| **PoG Certificate** | Issued: POGC-B9729A8723BD44D0 |

**VERDICT: FULL EXECUTION -- USD 50,000,000 RELEASED -- PoGC ISSUED**

### PATH C HALTED MID-EXECUTION -- Valid Start, Mandate Drift Detected

*The AI agent started with valid authority (88%). Mandate alignment degraded across turns -- automatic halt triggered.*

| | |
|---|---|
| **Authority level** | 88% -- VALID at start of execution |
| **Mandate score Turn 1** | N/A -- WARNING (below 0.65 threshold) |
| **Mandate score Turn 2** | N/A -- N/A (below 0.30 halt threshold) |
| **Behavioral chain** | CTCHC sealed as: HALTED -- forensic proof of mid-run halt |
| **Governance gate** | OSG (Settlement Gate) returned: REJECTED |
| **Funds released** | USD 0 -- halt triggered before settlement |

**VERDICT: EXECUTION HALTED MID-RUN -- USD 0 RELEASED -- MANDATE PROTECTED**

*Note: In all three paths, a Governance Contract (GCFR) was sealed with ML-DSA-65 BEFORE any action was taken -- capturing the agent identity, approved counterparties, mandate objectives, and freshness constraints at intake time. Admissible path contract: GCFR-D0BAF6C47CED42E8 (formed at 2026-05-30T20:59:02 UTC).*

## INDEPENDENT VERIFICATION GUIDE -- Any External Party Can Verify This

The verifier requires no OMNIX software, no OMNIX account, and no connection to any OMNIX server. It uses only two standard open-source Python libraries. The package is self-contained: the public key used to verify every signature is embedded inside the JSON file itself.

### Step 1 -- Install Two Standard Libraries

Open a terminal. Run:

```
pip install cryptography dilithium3
```

*Both are standard open-source libraries. No OMNIX code required.*

### Step 2 -- Download the Package and Verifier

Download both files from the OMNIX evidence repository. No build step, no installation, no dependencies beyond Step 1.

| | |
|---|---|
| **Package (JSON)** | OMNIX-RTE-001_20260530_205929.json |
| **Verifier (Python)** | verify_treasury_execution_trace.py |
| **Package ID** | OMNIX-RTE-001-694384C0419A4715 |

### Step 3 -- Run the Verifier

In the same directory as both files, run:

```
python verify_treasury_execution_trace.py \
    OMNIX-RTE-001_20260530_205929.json
```

### Step 4 -- Expected Output

A correctly signed, unmodified package produces:

```
TOTAL CHECKS  :  187
PASSED         :  186
FAILED         :  0
WARNINGS       :  1  (non-blocking -- DR TTL expired, signature still valid)

VERDICT: ALL VERIFICATIONS PASS
```

*The 1 warning is architectural: the Delegation Receipt on Path A carries a 21-minute TTL that expired after the package was generated. The ML-DSA-65 signature remains valid -- the warning is informational only.*

### Optional -- Verify Intake and Predicate Formation Only (IPFL Layer)

```
python verify_treasury_execution_trace.py \
    OMNIX-RTE-001_20260530_205929.json --verify-intake
```

*Expected: 44 / 44 PASS -- Intake Formation Report for all 3 paths.*

**Security Guarantee**

Every hash in the package is recomputed from scratch by the verifier. Every ML-DSA-65 signature is verified against the public key embedded in the package. Any tampering -- with any field, in any path, at any step -- causes at least one check to FAIL.

## VERIFICATION RESULTS -- 187 Checks | 0 Failures

| 187 | 186 | 0 | 1 |
|---|---|---|---|
| TOTAL CHECKS | PASSED | FAILED | WARNINGS |
| RFC-ATF-1 to RFC-ATF-6 | All cryptographic checks | Zero failures | Non-blocking / TTL |

**Intake & Predicate Formation (IPFL):** 44 / 44 -- Governance Contract sealed before Turn 0, all 3 paths

## What Was Verified

| | | |
|---|---|---|
| **Governance Contract (IPFL)** | **44 checks** | IAD, SAR, MFR, CPS, FPS hashes + PQC signatures per path |
| **Authority chain (DR + MBR)** | **18 checks** | Delegation Receipt integrity, MAR constraint, MBR binding |
| **Runtime continuity (RCR + MAS)** | **12 checks** | CES formula, band classification, Mandate Alignment Scores |
| **Counterfactual analysis (CGE)** | **14 checks** | CFR content hashes, CAT attestation, selected-path flag |
| **HALT evidence (Path A)** | **20 checks** | Refusal receipt, CTCHC seal, OSG REJECTED, no settlement |
| **Settlement evidence (Path B)** | **27 checks** | PoGC, MBR Seal, OSG APPROVED, XRPL/SWIFT refs, outcome |
| **Interrupted execution (Path C)** | **33 checks** | Turn-by-turn BAR/MAS, CTCHC HALTED, OSG REJECTED, no PoGC |
| **Post-execution forensics** | **14 checks** | TCS regulatory snapshot, replay proofs, offline verifiable flag |
| **Package structure** | **5 checks** | Package type, paths present, PQC key embedded, invariant count |

## Invariant Families Demonstrated -- 46 Total

| | | |
|---|---|---|
| **ATF** | Agent Trust Fabric (RFC-ATF-1) | **3 inv.** |
| **BEV** | Behavioral Execution Verification (RFC-ATF-6) | **7 inv.** |
| **CGE** | Counterfactual Governance Engine (RFC-ATF-5) | **4 inv.** |
| **IPFL** | Intake and Predicate Formation Layer (ADR-204) | **8 inv.** |
| **MIVP** | Mandate Integrity Verification Protocol (ADR-194) | **7 inv.** |
| **PoGR** | Proof of Governance Registry (ADR-186) | **3 inv.** |
| **RGC** | Runtime Governance Continuity (RFC-ATF-2) | **2 inv.** |
| **RTE** | Runtime Treasury Execution (ADR-201) | **11 inv.** |
| **TGB** | Temporal Governance Bridge (RFC-ATF-5) | **1 inv.** |

## Regulatory Alignment

| | |
|---|---|
| **EU AI Act Art. 9** | Risk management, human oversight, and audit trail requirements for high-risk AI systems |
| **MiCA Title VI** | Crypto-asset service provider governance and transaction integrity requirements |
| **DORA Art. 11** | Digital Operational Resilience Act -- ICT risk management and incident reporting |
| **FIPS 204** | Post-quantum cryptographic standard (ML-DSA-65 / Dilithium-3) -- NIST approved |

## PROOF OF GOVERNANCE CERTIFICATE -- The Auditable Record

A Proof of Governance Certificate (PoGC) is issued only when the AI agent completes execution with zero mandate violations, under valid authority, with every behavioral step recorded and sealed. It is the equivalent of a compliance receipt for AI-driven decisions -- verifiable offline, tamper-evident, and post-quantum signed.

PROOF OF GOVERNANCE CERTIFICATE

**POGC-B9729A8723BD44D0**

**CERTIFICATION TIER: MANDATE-BOUND -- 0 violations / 0 warnings across all turns**

### Certificate Fields

| | |
|---|---|
| **Certificate ID** | POGC-B9729A8723BD44D0 |
| **Session ID** | SESSION-CE41ACF073304B50 |
| **Agent** | OMNIX-AGENT-TREASURY-001 |
| **Issued at** | 2026-05-30 20:59 UTC |
| **Certification tier** | **MANDATE-BOUND** |
| **PQC algorithm** | ML-DSA-65 (FIPS 204 -- post-quantum standard) |
| **Regulatory tags** | EU AI Act Art. 9 / MiCA Title VI / DORA Art. 11 |
| **Standard** | RFC-ATF-6 (ADR-181/182/183/186) . ADR-194 (MIVP) |

### Mandate Integrity Seal -- Behavioral Record

| | |
|---|---|
| **Seal ID** | MBRSEAL-FAE259E5AF0948D9 |
| **Mandate certification** | N/A |
| **Total turns executed** | 4 |
| **Mandate violations** | 0 (MANDATE-BOUND requires 0) |
| **Algorithm** | ML-DSA-65 (Dilithium-3, FIPS 204) |

### Tamper-Evident Content Hash -- Verify Offline

SHA3-256 hash of all certificate fields (excluding signature). Recomputable from the package JSON:

`3bb415a1ca3190dfb745de076006b0964152a0d53799cfe696db7f2e13f78c84`

Mandate Integrity Seal hash (covers all behavioral turns):

`d0a22a49cf8e3669069fc2cbca59eb9644535de29457f571ff3ee76b925edda6`

**ATTESTATION**

This certificate is embedded in the verifiable package OMNIX-RTE-001-694384C0419A4715. Every field is covered by a post-quantum ML-DSA-65 signature verifiable offline using only the public key embedded in the same package. Issued under RFC-ATF-6 (ADR-181/182/183/186) and ADR-194 (MIVP). OMNIX QUANTUM LTD | Harold Nunes, Founder.